AP Precalculus

Unit 3, Lesson 3.7

Exit Ticket — Answer Key

Name: ____________________________ Date: ________________ Period: _________

Exit Ticket — Key

The table below gives values of a periodic function f over one period.

x	0	2	4	6	8	10
$f(x)$	3	11	3	-5	3	11

1. Identify the maximum and minimum values of f.
Maximum = **11** Minimum = -5

2. Compute the amplitude and midline.

$$A = \frac{\max - \min}{2} = \frac{11 - (-5)}{2} = \frac{16}{2} = 8$$

$$D = \frac{\max + \min}{2} = \frac{11 + (-5)}{2} = \frac{6}{2} = 3$$

3. Estimate the period k from the table. Explain how you found it.
k = **8** Explanation: **Consecutive maxima occur at $x = 2$ and $x = 10$; the distance is $10 - 2 = 8$.**

4. Using your period, compute $B = \dfrac{2\pi}{k} = \dfrac{2\pi}{8} = \dfrac{\pi}{4}$

5. A student uses the model $f(x) = 8\cos\left(\frac{\pi}{4}(x-2)\right) + 3$ to predict $f(3)$. Is $x = 3$ inside the contextual domain for this data set? Circle one: **YES** / NO

Explain: **The data set covers $0 \le x \le 10$ (one period). $x = 3$ falls within this interval, so it is inside the contextual domain.**